

Three-Body Coupling Test

Three bodies with the same density contrast (+0.3 g/cc) and different susceptibility:
A strongly magnetic (0.05 SI), B weakly (0.01 SI), C not at all. Can a joint inversion tell them apart, and which coupling of the density and susceptibility models helps?
Six couplings are compared on the same data, first with block-shaped bodies, then with dipping intrusions.

Set-up

DATA	MESH	REGULARIZATION	COUPLING WEIGHT
gz and TMI at 357 stations (75 m grid), vertical inducing field, 2 % noise of each peak	50 m tensor cells, core to 600 m depth, 600 m padding; data modelled on a finer 25 m mesh	L1–L2 elastic net per model (IRLS up to 40 cycles, L1 share 0.8, sensitivity weighting); PGI and the group lasso bring their own	1 for every weighted coupling: as strong as the regularization (unit-free)
BLOCKS	DIPPING INTRUSIONS		
200 × 300 × 200 m, 100–300 m deep, centred at x = 300, 750, 1200 m	120 m wide, dipping 55° east, from 75 to 375 m depth		

No coupling

reference

Both models inverted together with one β , nothing linking them.

Linear correspondence

petrophysical · SimPEG

One relation in every cell, here $\Delta\rho = 6\chi$ (body A's ratio): a deliberately partial rule.

Cross-gradient

structural · Gallardo & Meju 2003

The models' gradients are parallel: boundaries coincide, values stay free.

PGI

petrophysical · Astic & Oldenburg 2019

Each cell belongs to one rock unit; given the three true units, the best case for it.

Joint total variation

structural · Haber & Holtzman Gazit 2013

The models change in the same places and are flat elsewhere.

Group lasso

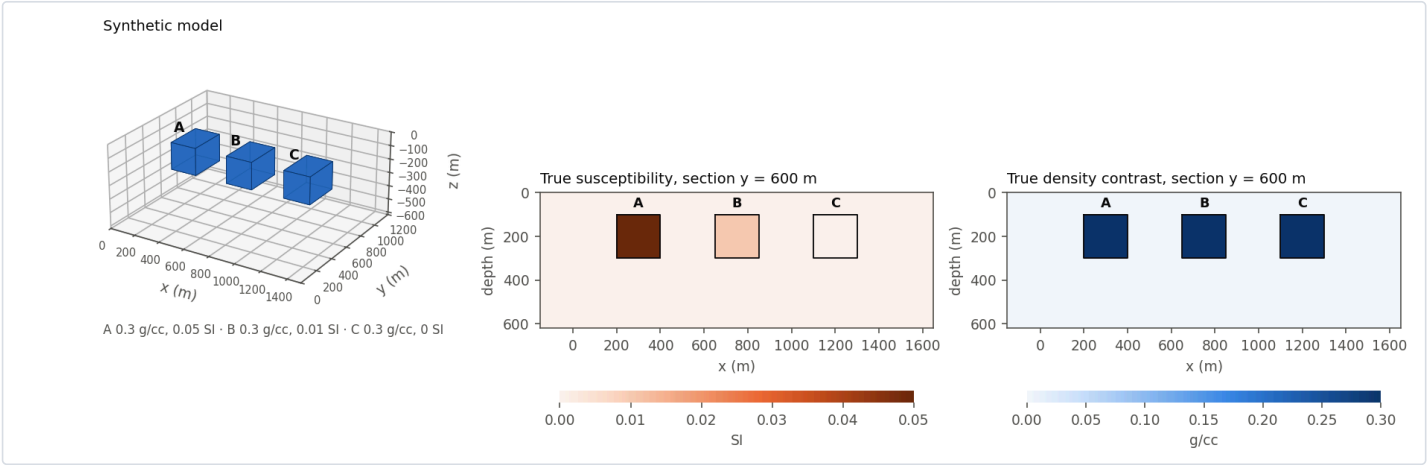
joint sparsity · Utsugi 2025

Anomalies share their cells, with any sign or ratio; λ_1 from the L-curve, $\lambda_2 = 0.3$.

Blocks

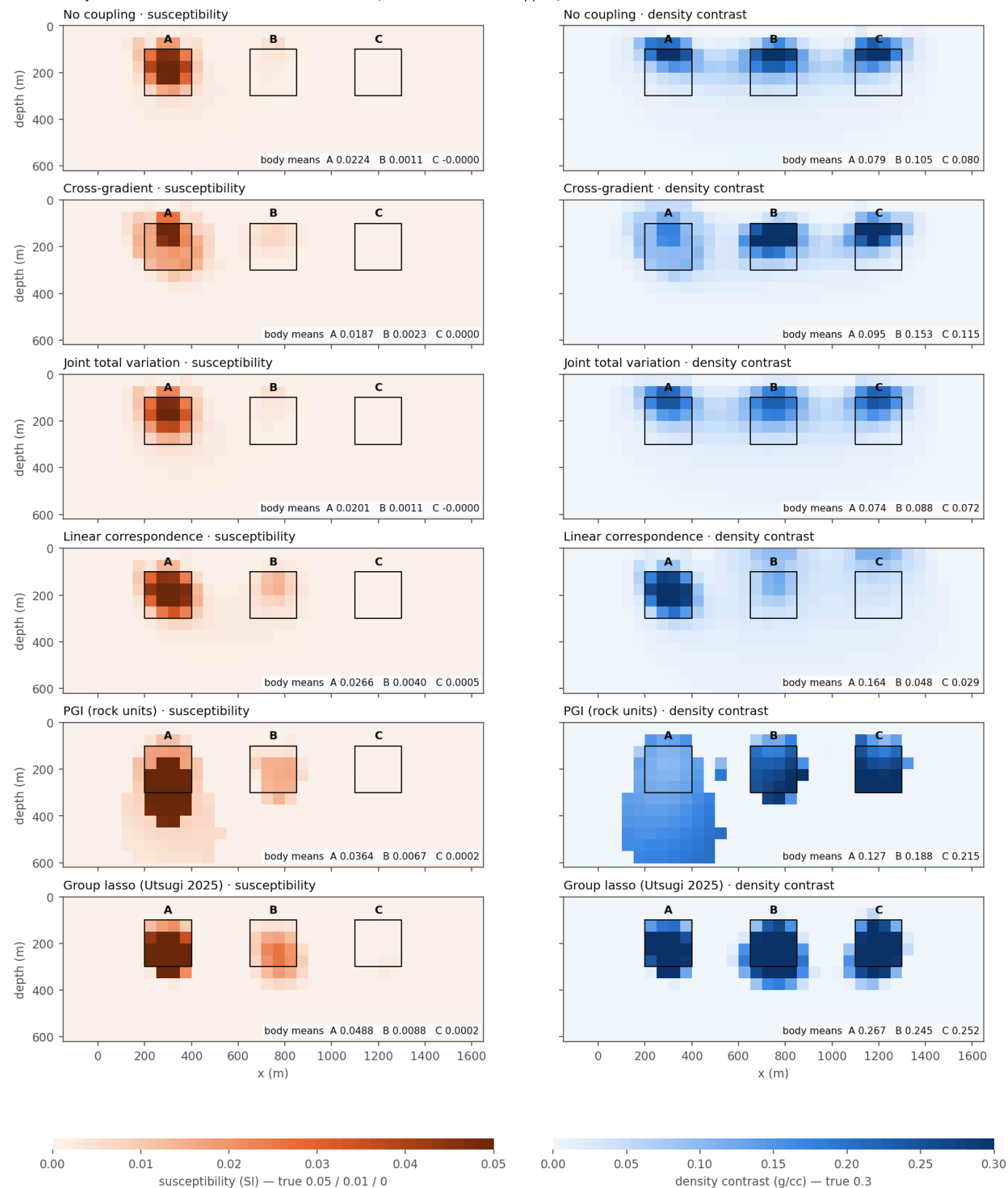
Three boxes at the same depth. The anomalies overlap little, so this is the easier case.

[Open the interactive workflow \(https://claude.ai/artifact/XzpkrmuoTQtn2VVzoY4vGb\)](https://claude.ai/artifact/XzpkrmuoTQtn2VVzoY4vGb) 3D models against the true model, every run's data fit and settings



The synthetic model and its true sections at y = 600 m. All three bodies have the same density contrast (+0.3 g/cc); only their susceptibility differs.

Sections at $y = 600$ m on the true model's colour scales (values above them clipped)

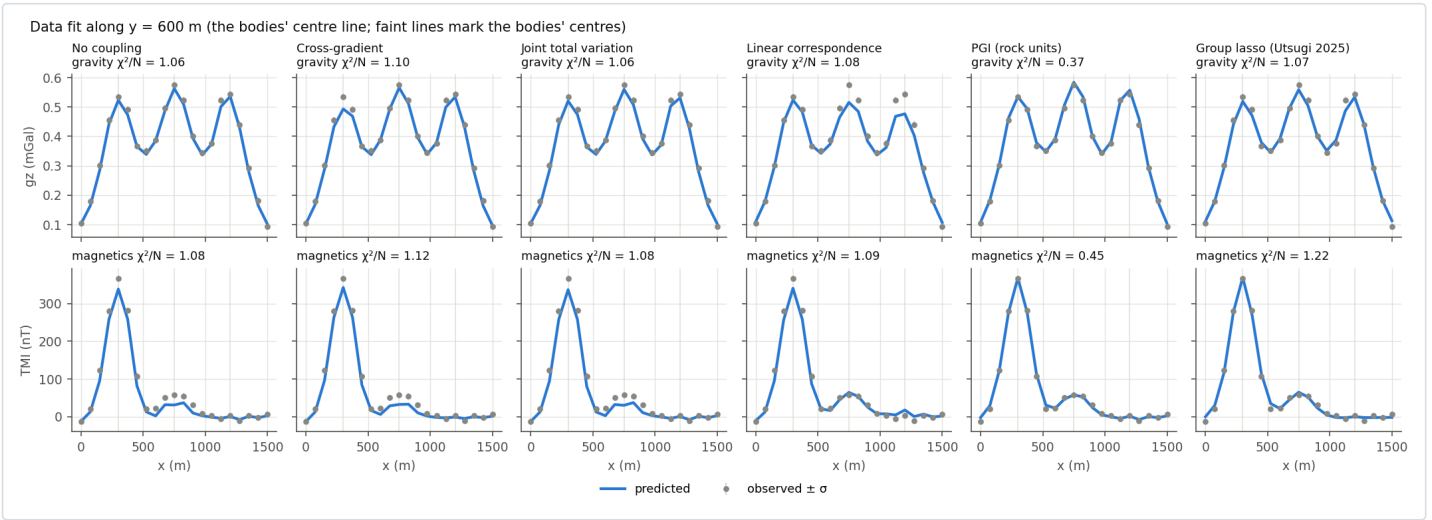


Each coupling's susceptibility (left) and density contrast (right) on the section through the bodies, on the true model's colour scales. Outlines mark the true bodies.

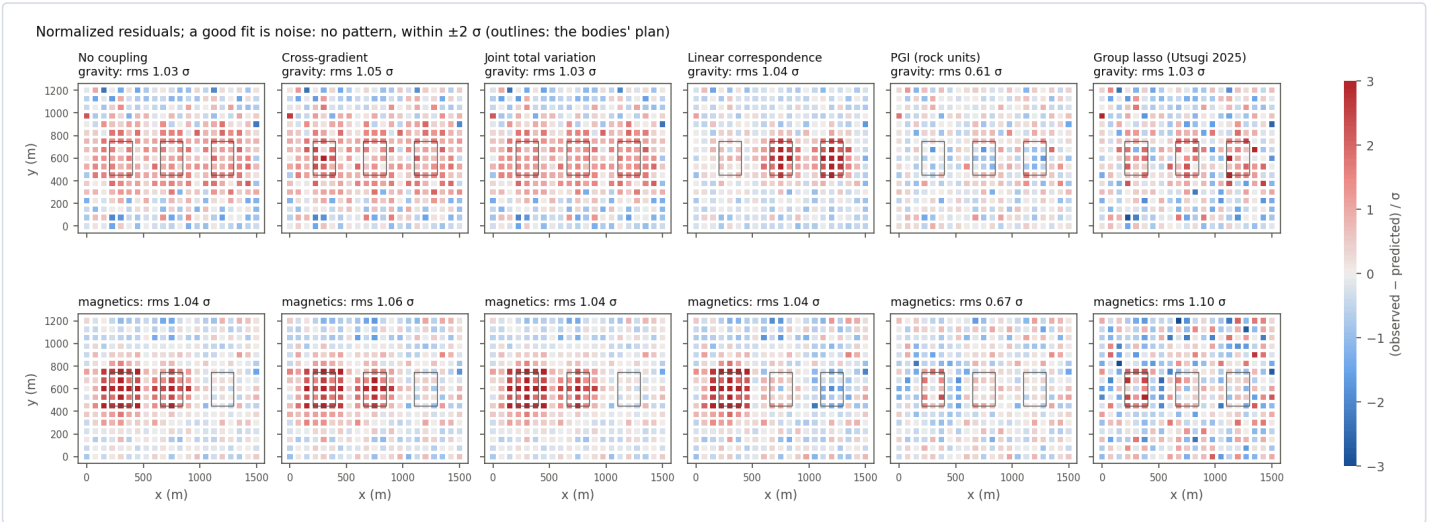
Recovered values and fit

Coupling	Body A		Body B		Body C		$\Delta\rho$ rms error	χ rms error	$\Delta\rho$ outside bodies	χ^2/N grav / mag	s
	$\Delta\rho$	χ	$\Delta\rho$	χ	$\Delta\rho$	χ					
True model	0.300	0.0500	0.300	0.0100	0.300	0.0000	0	0	0	1	
No coupling	0.079	0.0224	0.105	0.0011	0.080	-0.0000	0.0423	0.0034	0.0054	1.06 / 1.08	39
Cross-gradient	0.095	0.0187	0.153	0.0023	0.115	0.0000	0.0380	0.0036	0.0044	1.10 / 1.12	80
Joint total variation	0.074	0.0201	0.088	0.0011	0.072	-0.0000	0.0421	0.0035	0.0057	1.06 / 1.08	66
Linear correspondence	0.164	0.0266	0.048	0.0040	0.029	0.0005	0.0420	0.0031	0.0062	1.08 / 1.09	64
PGI	0.127	0.0364	0.188	0.0067	0.215	0.0002	0.0436	0.0042	0.0092	0.37 / 0.45	40
Group lasso	0.267	0.0488	0.245	0.0088	0.252	0.0002	0.0342	0.0044	0.0015	1.07 / 1.22	181

Body means of the inverted models ($\Delta\rho$ in g/cc, χ in SI) over the cells inside each body; errors are the rms difference from the true model over the core of the mesh; the lowest of each is in bold. χ^2/N near 1 fits the data to their noise. PGI put 48/96 of A's cells in unit A, 40/96 of B's in B, 70/96 of C's in C, and called 384 cells outside the bodies a body.



Observed data (grey, $\pm\sigma$) and each coupling's predicted data (blue) along $y = 600$ m.



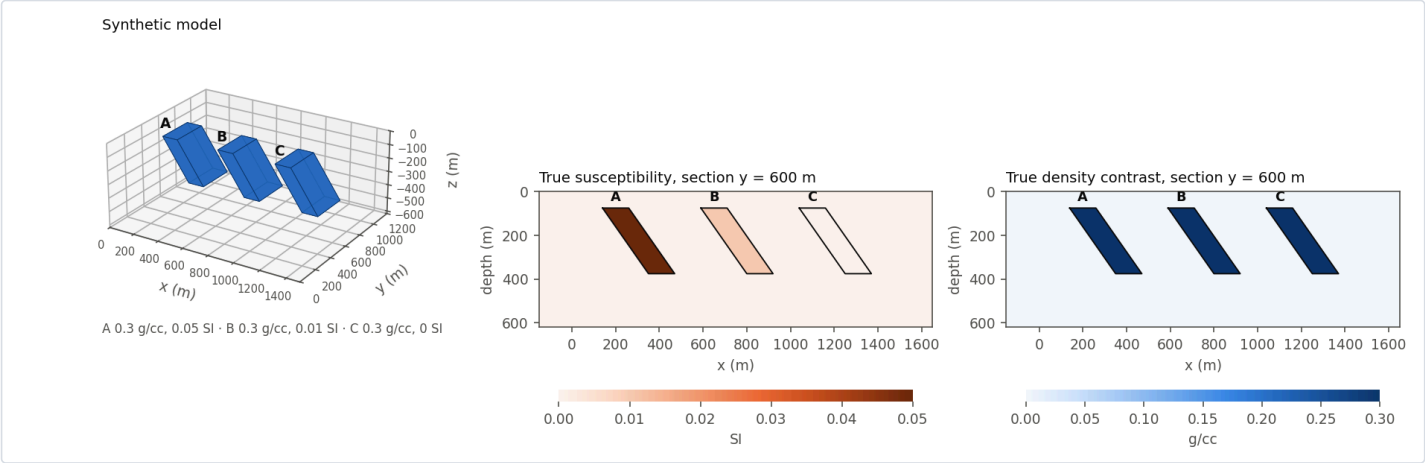
Residuals divided by the data error. A fit that has explained the bodies leaves noise; red or blue patches over a body mean part of its anomaly is not explained.

Blocks. Without coupling, L1–L2 recovers each body at about a third of its density contrast (0.08–0.11 of 0.3) and B's weak susceptibility barely at all (0.0011 of 0.01). The cross-gradient raises the densities to 0.10–0.15; the joint total variation leaves them about where they were. The group lasso recovers 80–90 % of every density contrast, both susceptibilities (0.049 and 0.009) and none in C, with the least density outside the bodies. The linear correspondence, told $\Delta\rho = 6\chi$ everywhere, lowers C's density to 0.029 and gives it susceptibility it does not have; the magnetic residual map shows the misfit over C. PGI, given the exact rock units, raises the densities to 0.13–0.22 but fits the data to χ^2/N 0.37 / 0.45 and puts a false dense zone under A.

Dipping intrusions

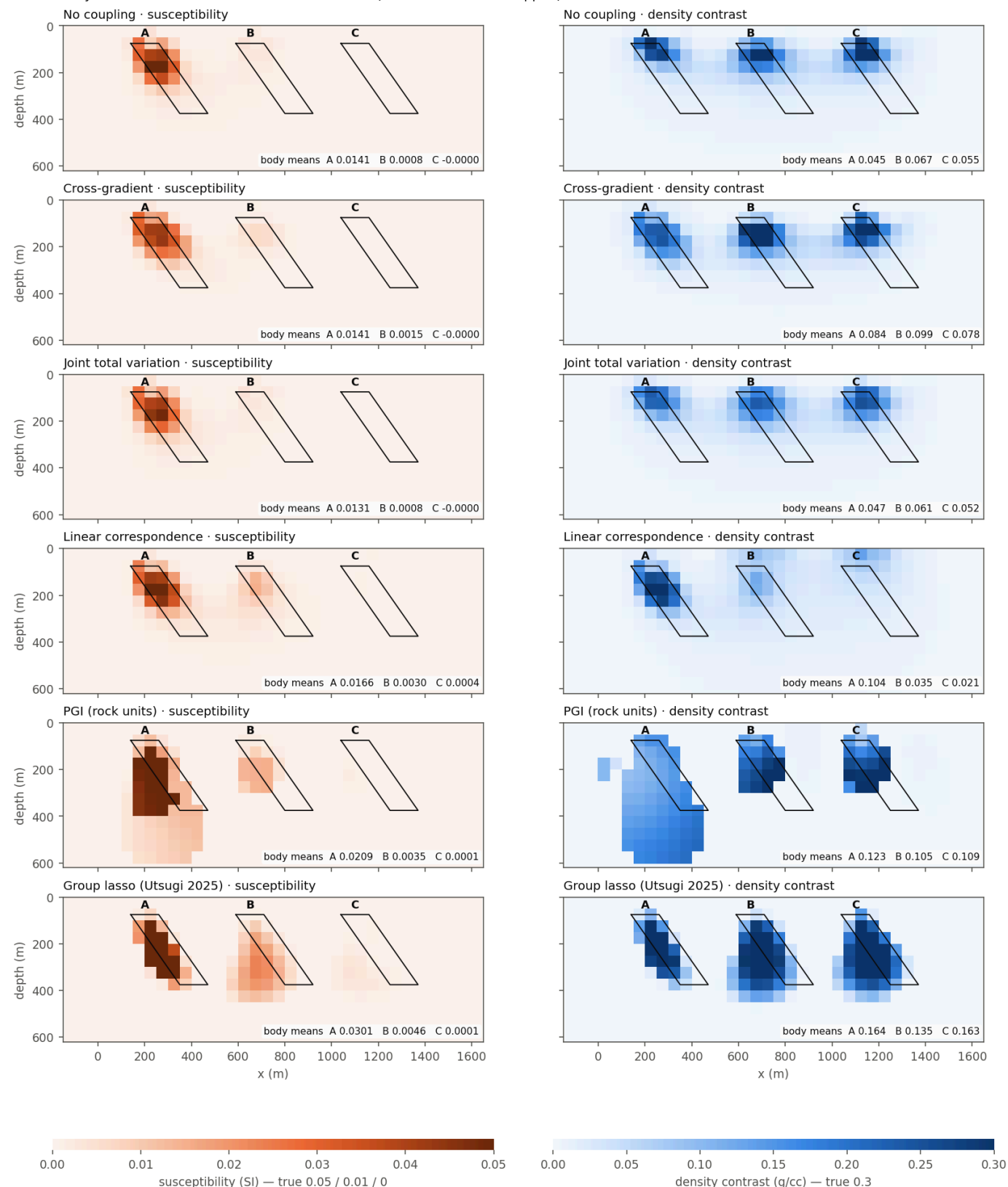
The same three bodies as tabular intrusions dipping 55° to the east, reaching deeper.

[Open the interactive workflow \(https://claude.ai/artifact/Rv224oiaVHMcvRQfkCsTyQ\)](https://claude.ai/artifact/Rv224oiaVHMcvRQfkCsTyQ) 3D models against the true model, every run's data fit and settings



The synthetic model and its true sections at y = 600 m. All three bodies have the same density contrast (+0.3 g/cc); only their susceptibility differs.

Sections at $y = 600$ m on the true model's colour scales (values above them clipped)

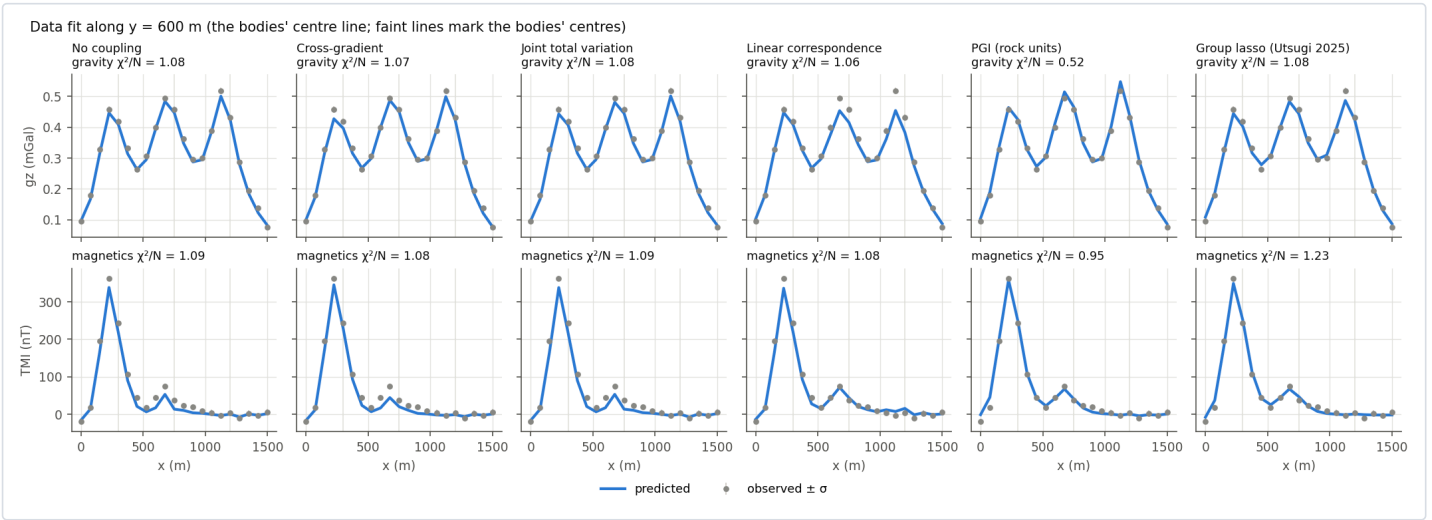


Each coupling's susceptibility (left) and density contrast (right) on the section through the bodies, on the true model's colour scales. Outlines mark the true bodies.

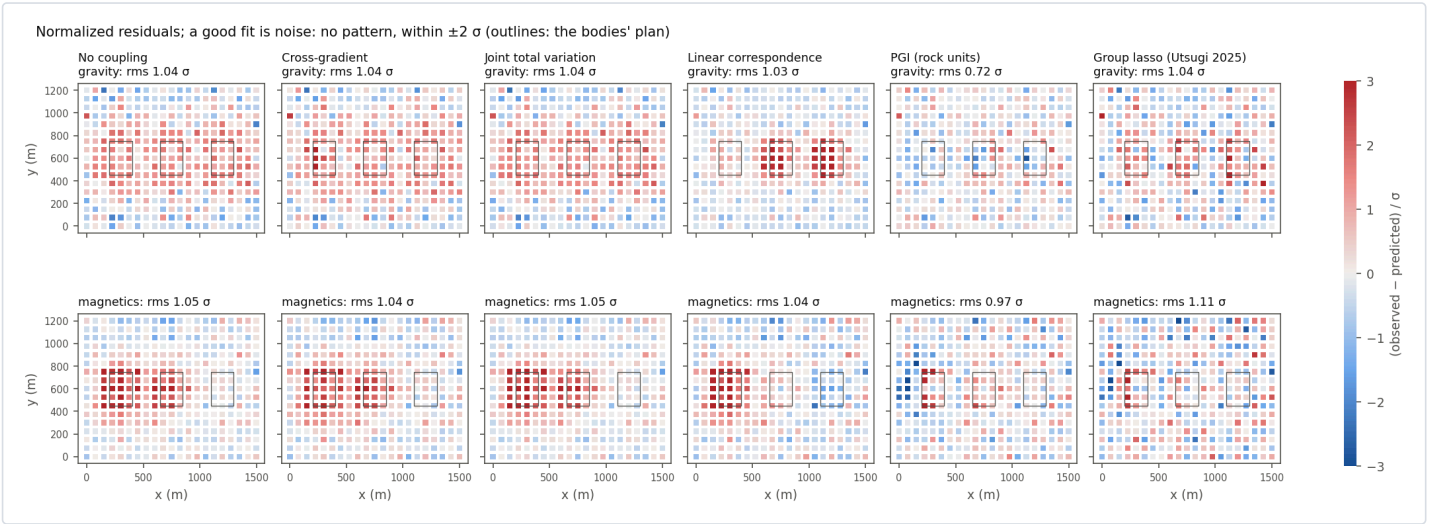
Recovered values and fit

Coupling	Body A		Body B		Body C		$\Delta\rho$ rms error	χ rms error	$\Delta\rho$ outside bodies	χ^2/N grav / mag	s
	$\Delta\rho$	χ	$\Delta\rho$	χ	$\Delta\rho$	χ					
True model	0.300	0.0500	0.300	0.0100	0.300	0.0000	0	0	0		1
No coupling	0.045	0.0141	0.067	0.0008	0.055	-0.0000	0.0441	0.0039	0.0053	1.08 / 1.09	43
Cross-gradient	0.084	0.0141	0.099	0.0015	0.078	-0.0000	0.0412	0.0039	0.0046	1.07 / 1.08	110
Joint total variation	0.047	0.0131	0.061	0.0008	0.052	-0.0000	0.0434	0.0039	0.0054	1.08 / 1.09	106
Linear correspondence	0.104	0.0166	0.035	0.0030	0.021	0.0004	0.0427	0.0037	0.0058	1.06 / 1.08	107
PGI	0.123	0.0209	0.105	0.0035	0.109	0.0001	0.0493	0.0050	0.0091	0.52 / 0.95	30
Group lasso	0.164	0.0301	0.135	0.0046	0.163	0.0001	0.0451	0.0048	0.0040	1.08 / 1.23	222

Body means of the inverted models ($\Delta\rho$ in g/cc, χ in SI) over the cells inside each body; errors are the rms difference from the true model over the core of the mesh; the lowest of each is in bold. χ^2/N near 1 fits the data to their noise. PGI put 18/84 of A's cells in unit A, 19/84 of B's in B, 33/84 of C's in C, and called 367 cells outside the bodies a body.



Observed data (grey, $\pm\sigma$) and each coupling's predicted data (blue) along $y = 600$ m.



Residuals divided by the data error. A fit that has explained the bodies leaves noise; red or blue patches over a body mean part of its anomaly is not explained.

Dipping intrusions. Every method loses amplitude and some of the dip: the bodies reach 375 m and the data resolve their tops best. The group lasso still gives the highest densities (0.14–0.16 of 0.3) and follows A's dip in susceptibility; the cross-gradient raises the L1–L2 densities from 0.05–0.07 to 0.08–0.10. The linear correspondence again drags C's density down. PGI classifies fewer cells correctly than for the blocks (A 18, B 19, C 33 of 84).

The group lasso, taken apart

The group lasso is an L1 + L2 regularization in which the L1 term acts on each cell's pair of values (density, susceptibility) instead of on each value alone. The pairing is the coupling. Three runs show what comes from where:

- 1. **L1–L2 solved by IRLS, no coupling.** The "No coupling" run above, and the base of the cross-gradient, joint total variation and linear correspondence runs.
- 2. **L1 + L2 solved by ADMM, no coupling.** An ordinary L1 on every value, with the same weighting, $\lambda_2 = 0.3$, L-curve choice of λ_1 and solver as run 3. It is not a group lasso: nothing links the models.
- 3. **Group lasso, solved by ADMM.** Run 2 with the L1 applied to each cell's pair.

RUN 2: L1 + L2	$\frac{1}{2}\ \mathbf{b} - \mathbf{Z}\boldsymbol{\zeta}\ ^2 + \lambda_1 \sum_k (\Delta\rho_k + \chi_k) + \frac{1}{2}\lambda_2\ \boldsymbol{\zeta}\ ^2$
RUN 3: GROUP LASSO	$\frac{1}{2}\ \mathbf{b} - \mathbf{Z}\boldsymbol{\zeta}\ ^2 + \lambda_1 \sum_k \sqrt{(\Delta\rho_k^2 + \chi_k^2)} + \frac{1}{2}\lambda_2\ \boldsymbol{\zeta}\ ^2$

Runs 1 and 2 have the same kind of regularization and differ in how it is solved; runs 2 and 3 are solved the same way and differ only in the pairing. Since the gravity data depend only on $\Delta\rho$ and the magnetic data only on χ , run 2 is two independent inversions; only the square root in run 3 ties the models together.

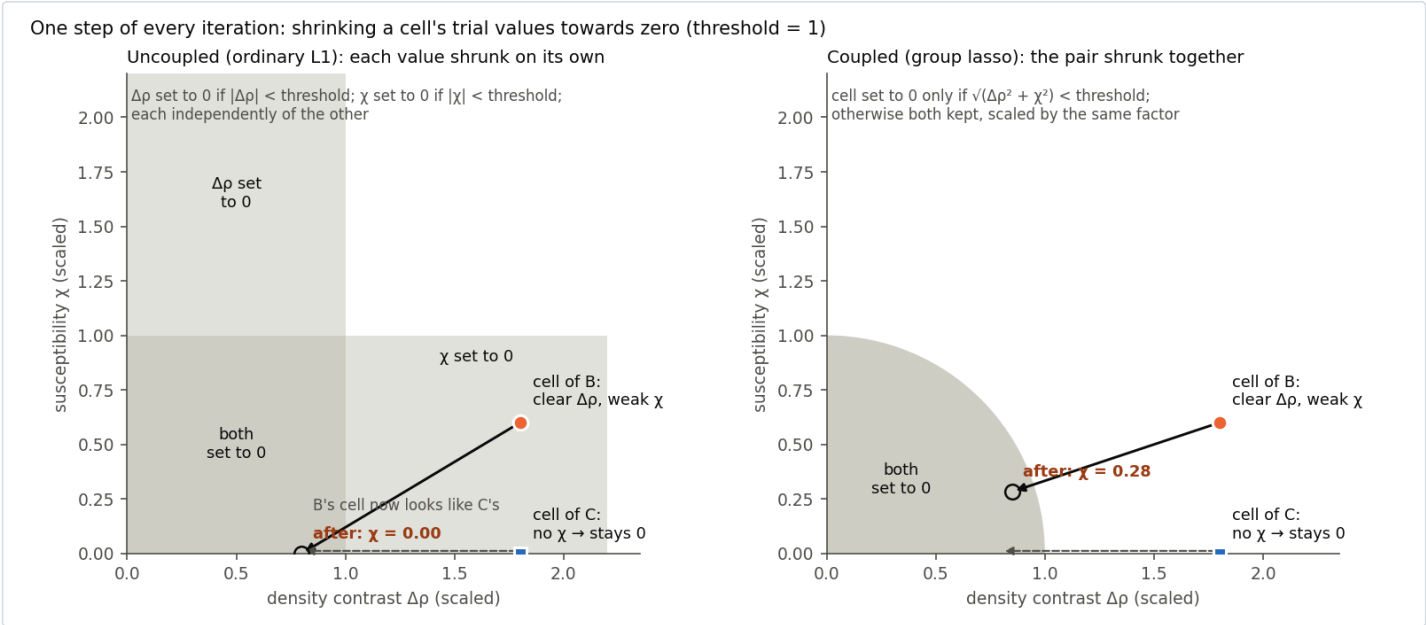
Runs 2 and 3: what the pairing does, in one step of the solver

Every iteration of the ADMM solver does two things:

1. From the data, it computes a trial value of every cell's density and susceptibility.
2. It shrinks these values towards zero; a value below a threshold becomes exactly zero. This step is what makes the models compact.

Runs 2 and 3 differ only in step 2:

- **Run 2, L1 + L2:** the density and the susceptibility of a cell are compared with the threshold separately. Either one is set to zero if it is small, whatever the other value of the same cell is.
- **Run 3, group lasso:** the two values of a cell are compared with the threshold together, by their joint size $\sqrt{(\Delta\rho^2 + \chi^2)}$. The cell is set to zero only if both are small; otherwise both are kept and scaled down by the same factor.



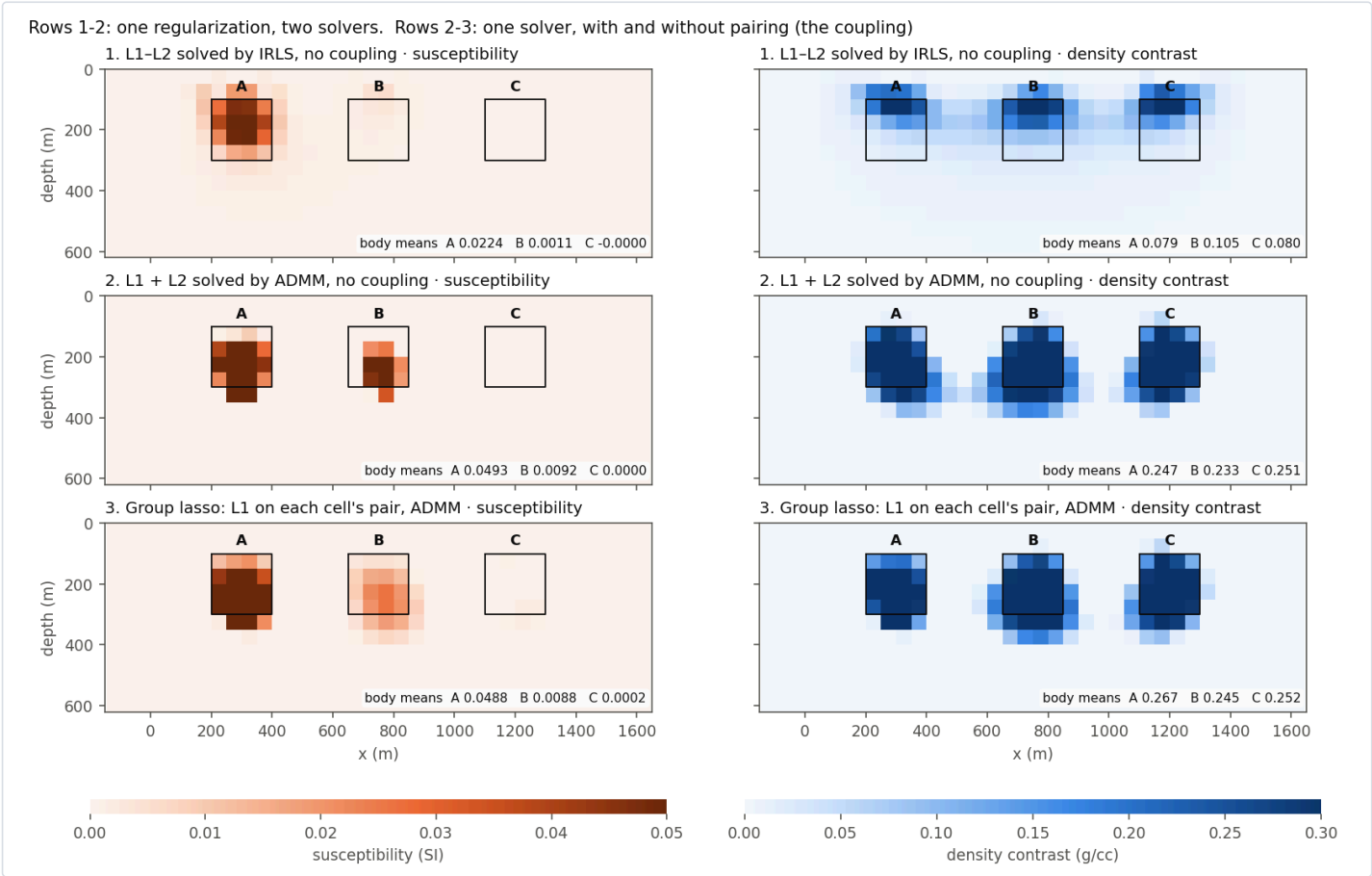
The one step in which runs 2 and 3 differ, drawn for a threshold of 1 in the solver's scaled units (an illustration, not data). Grey: the trial values that are set to zero.

A cell of body B has a clear density and a weak susceptibility. In run 2 its weak susceptibility falls below the threshold and becomes zero, so the cell ends up looking like a cell of C. B's magnetic anomaly still has to be explained, and the few cells whose susceptibility does pass the threshold take it all: they end up at eight times the true value. In run 3 the cell is kept because of its density, and its susceptibility is kept with it: B's susceptibility spreads over its dense cells at about the right level.

A cell of body C has density but no magnetic signal, so its trial susceptibility is about zero in both runs and stays zero. The pairing does not create magnetization; it lets a weak one survive in cells that are active anyway.

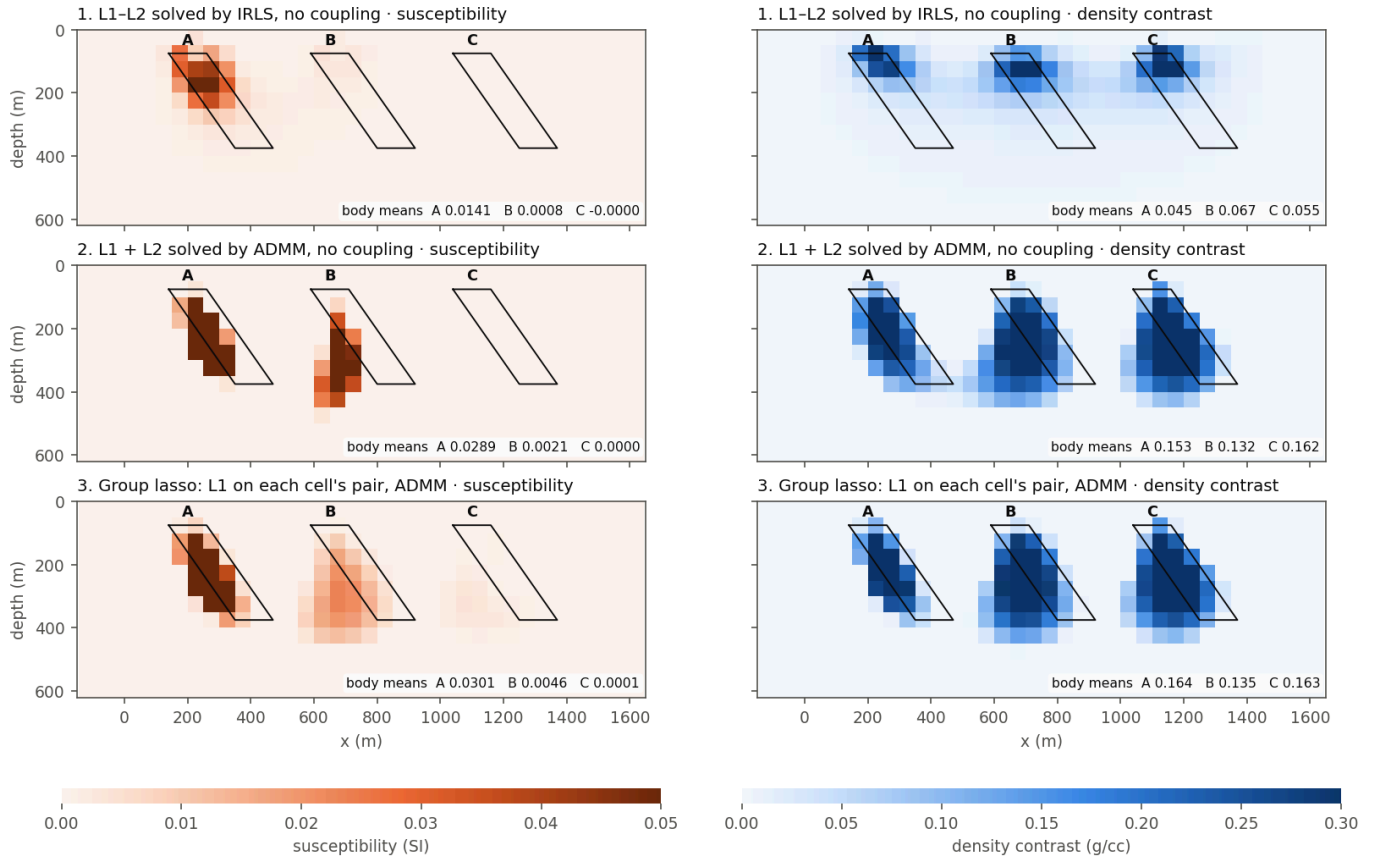
The three runs side by side

Shape	Run	$\Delta\rho$ A	$\Delta\rho$ B	$\Delta\rho$ C	χ A	χ B	χ C	χ^2/N grav / mag
True		0.300	0.300	0.300	0.0500	0.0100	0.0000	1
Blocks	1. L1–L2, IRLS, no coupling	0.079	0.105	0.080	0.0224	0.0011	-0.0000	1.06 / 1.08
Blocks	2. L1 + L2, ADMM, no coupling	0.247	0.233	0.251	0.0493	0.0092	0.0000	1.12 / 1.36
Blocks	3. Group lasso, ADMM	0.267	0.245	0.252	0.0488	0.0088	0.0002	1.07 / 1.22
Dipping	1. L1–L2, IRLS, no coupling	0.045	0.067	0.055	0.0141	0.0008	-0.0000	1.08 / 1.09
Dipping	2. L1 + L2, ADMM, no coupling	0.153	0.132	0.162	0.0289	0.0021	0.0000	1.12 / 1.34
Dipping	3. Group lasso, ADMM	0.164	0.135	0.163	0.0301	0.0046	0.0001	1.08 / 1.23



Blocks. Row 1: L1–L2 solved by IRLS, no coupling. Row 2: L1 + L2 solved by ADMM, no coupling. Row 3: the group lasso (L1 on each cell's pair), solved by ADMM.

Rows 1-2: one regularization, two solvers. Rows 2-3: one solver, with and without pairing (the coupling)



Dipping intrusions: the same three runs.

Runs 1 and 2: the solver. With the same kind of regularization and no coupling, the ADMM solution recovers 0.23–0.25 g/cc in the blocks against 0.08–0.11 for IRLS, and C, which has no magnetic signal to borrow, reaches 0.25 as well, so this gain does not come from coupling. ADMM finds the exact sparse solution (cells set exactly to zero), the sensitivity weighting $1/\|k_j\|$ offsets the decay of the kernels with depth, and λ_1 comes from the L-curve; IRLS approximates the L1 norm by iterative reweighting with a smoothing threshold and weights the cells differently.

Runs 2 and 3: the pairing, which is the coupling. The effect described above, measured: in run 2, B's susceptibility survives in only a few cells, pushed to 0.083 SI (the truth is 0.01), and 60 % of B's dense cells have none. In run 3 it fills 95 % of them with a peak of 0.027 (dipping bodies: 18 % → 84 % of the dense cells, peak 0.038 → 0.018). The magnetic fit improves (χ^2/N 1.22 against 1.36) and the densities gain up to 0.02. The pairing suits bodies whose magnetic parts are also dense, as here; it does not force a dense cell to be magnetic (C keeps $\chi \approx 0$) or the two properties to share a ratio or sign.

So each coupling is best judged against its own uncoupled run: the group lasso against run 2; the cross-gradient, the joint total variation and the linear correspondence against run 1.

A note on the L1–L2 runs: reweighting cycles

With 20 IRLS cycles every L1–L2 run stopped at the limit before converging. With 40, the uncoupled run and the joint total variation converge and their densities rise by 60–85 %; the cross-gradient gains most from the extra cycles, and it and the linear correspondence still run to the limit, so more cycles may raise them further. All results above use 40 cycles.

Shape	Coupling	$\Delta\rho$ A / B / C, 20 IRLS cycles	$\Delta\rho$ A / B / C, 40 cycles	40 cycles ended by
Blocks	No coupling	0.046 / 0.059 / 0.047	0.079 / 0.105 / 0.080	converged
Blocks	Cross-gradient	0.067 / 0.062 / 0.049	0.095 / 0.153 / 0.115	40-cycle limit
Blocks	Joint total variation	0.045 / 0.053 / 0.045	0.074 / 0.088 / 0.072	converged
Blocks	Linear correspondence	0.088 / 0.035 / 0.029	0.164 / 0.048 / 0.029	40-cycle limit
Dipping	No coupling	0.028 / 0.037 / 0.030	0.045 / 0.067 / 0.055	converged
Dipping	Cross-gradient	0.046 / 0.039 / 0.031	0.084 / 0.099 / 0.078	40-cycle limit
Dipping	Joint total variation	0.029 / 0.034 / 0.029	0.047 / 0.061 / 0.052	converged
Dipping	Linear correspondence	0.056 / 0.026 / 0.020	0.104 / 0.035 / 0.021	40-cycle limit

What the couplings did

- **Group lasso** recovered the most: densities close to the truth, the weak susceptibility of B in the right cells, none in C, and the least structure outside the bodies. Its L1 + L2 solved by ADMM gives the compact, strong bodies (an uncoupled L1 + L2 solved the same way does nearly as well on density); its pairing, the coupling, puts B's weak susceptibility in B's dense cells and improves the fit. It is the slowest here (3–4 min).
- **Cross-gradient** improved clearly once the reweighting had enough cycles: densities 20–90 % higher than the uncoupled L1–L2 run, the largest gain among the couplings added to L1–L2.
- **Joint total variation** changed little at weight 1: its densities stayed slightly below the uncoupled run's.
- **Linear correspondence** is only as good as its relation: a single $\Delta\rho$ – χ line cannot describe three bodies with three different ratios, and C (dense, not magnetic) is the casualty.
- **PGI** sharpened the bodies when told the exact units, but overfitted the data and misplaced dense cells below A. With real units known only roughly, expect less.

One synthetic, one noise realization, one weight per coupling; timings on a laptop. Scripts: `examples/output/coupling_comparison/` (`run_coupling_comparison.py`, `make_report_figures.py`, `make_why_figure.py`, `make_report_html.py`). Couplings: `docs/joint_couplings.md`.